

D2 Report

Data Structure Similarity

The ContactBook also represents a tree structure. The original ContactBook acts as the root, while each recursed_book created after a collision acts as a child at a deeper level. The depth of the tree corresponds to the character position used to hash the nickname. Nicknames that share the same characters remain on the same path, while different characters place them into different branches. Therefore, the recursive ContactBooks form a hierarchical tree in which nicknames are partially organised according to their characters.

Testing Evidence

/Users/yihaoyu/Desktop/FIT2085/Ontrack_Files/D2/main

1. Add a contact
2. Retrieve a contact
3. Remove a contact
4. Search by prefix
5. Show the number of contacts
6. See the menu again
7. Quit

Enter command (1-7):

1

Enter the nickname:

alex

Enter the display name:

Alex

Enter the user score:

100

Added Alex under the nickname 'alex'

Enter command (1-7):

1

Enter the nickname:

alicia

Enter the display name:

Alicia

Enter the user score:

90

Added Alicia under the nickname 'alicia'

Enter command (1-7):

1

Enter the nickname:

bob

Enter the display name:

Bob

Enter the user score:

80

```
Added Bob under the nickname 'bob'
Enter command (1-7):
5
The contact book holds 3 contact(s)
Enter command (1-7):
2
Enter the nickname to retrieve:
alex
Nickname: alex | Display name: Alex | Score: 100
Enter command (1-7):
2
Enter the nickname to retrieve:
charlie
No contact is saved under 'charlie'
Enter command (1-7):
4
Enter the prefix to search for:
a
Yes - at least one contact has the prefix 'a'
Enter command (1-7):
4
Enter the prefix to search for:
al
Yes - at least one contact has the prefix 'al'
Enter command (1-7):
4
Enter the prefix to search for:
alex
Yes - at least one contact has the prefix 'alex'
Enter command (1-7):
4
Enter the prefix to search for:
ali
Yes - at least one contact has the prefix 'ali'
```

```
Enter command (1-7):
4
Enter the prefix to search for:
alice
No contact has the prefix 'alice'
Enter command (1-7):
4
Enter the prefix to search for:
bo
Yes - at least one contact has the prefix 'bo'
Enter command (1-7):
3
Enter the nickname to remove:
alicia
Removed the contact 'alicia'
Enter command (1-7):
5
The contact book holds 2 contact(s)
Enter command (1-7):
2
Enter the nickname to retrieve:
alicia
No contact is saved under 'alicia'
Enter command (1-7):
4
Enter the prefix to search for:
ali
No contact has the prefix 'ali'
```

Word Count